

Mathematics 2, part 4, lecture 3

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- ▶ Linear programming
 - ▶ ~~LP toy problem~~
 - ▶ ~~Problems with LP problems~~
 - ▶ ~~LP duality~~
 - ▶ ~~Dimension analysis~~
 - ▶ ~~LP interior point methods, why?~~
 - ▶ ~~GD in interior point methods~~
 - ▶ ~~When to stop descending?~~
 - ▶ ~~How to start?~~
 - ▶ ~~Time complexity.~~
 - ▶ Concluding remarks.

Initial solution (traditional)

$$\min 480x_1 + 600x_2$$

$$2x_1 + 3x_2 \geq 3$$

$$2x_1 + 1x_2 \geq 2$$

$$x_1, x_2 \geq 0$$

$$\min 480x_1 + 600x_2$$

$$2x_1 + 3x_2 \geq 3 - x_0$$

$$2x_1 + 1x_2 \geq 2 - x_0$$

$$x_1, x_2, x_0 \geq 0$$

$$\min x_0$$

$$2x_1 + 3x_2 \geq 3 - x_0$$

$$2x_1 + 1x_2 \geq 2 - x_0$$

$$x_1, x_2, x_0 \geq 0$$

$$\min 480x_1 + 600x_2 + Mx_0$$

$$2x_1 + 3x_2 \geq 3 - x_0$$

$$2x_1 + 1x_2 \geq 2 - x_0$$

$$x_1, x_2, x_0 \geq 0$$

Initial solution

We want to transform (P) — by possibly adding additional variables — to the *artificial problem* (art P) so that

- ▶ the feasible domain is bounded (preserving the optimum if it exists),
- ▶ flagging the possible infeasible problem,
- ▶ so that it is possible to find strictly feasible solutions to (art P) and its dual (D),
- ▶ which also satisfy the invariant (I3) for a suitable initial choice of μ_{init} .

Final rounding (recall)

Lemma

Let (x, y, s) and μ satisfy the invariants (I1), (I2), and (I3). Let x^* be an optimal solution to (P) and let (y^*, s^*) be an optimal solution to (D). Assume that the smallest nonzero value of x^* and s^* is at least L .

- ▶ If $x_i < L/(4m)$ then $x_i^* = 0$ in every optimal solution and
- ▶ if $s_i < L/(4m)$ then $s_i^* = 0$ in every optimal solution.

Theorem (strong duality)

For every i we have $x_i^* = 0$ in every optimal solution or $s_i^* = 0$ in every optimal solution. Thus

$$c^T x^* - b^T y^* = (x^*)^T s^* = 0.$$

Complexity of IP

Theorem

The interior point method (as described during lectures) terminates in polynomial number of steps.

LP, historical notes

- ▶ mid 19th century, Fourier: elimination of variables from a system of linear inequalities.
- ▶ 1940s, Dantzig: simplex algorithm (worst case exponential)
- ▶ 1940s, von Neuman: duality
- ▶ 1970s: ellipsoid method, first polynomial algorithm (not practical)
- ▶ 1984, Karmarkar: first interior-point algorithm (polynomial, practical)
- ▶ still open: is there a **strongly polynomial** algorithm for LP.

Bottom line: interior point methods work well for millions of variables+constraints, often with a fixed number (less than 100) iterations.

Interior point methods not just for linear programming

► quadratic programming

$$\begin{aligned} \min \quad & x^T Q^T Q x + c^T x \\ & Ax \leq b \end{aligned}$$

► semidefinite programming

$$\begin{aligned} \min \quad & \text{trace}(C^T X) \\ & \text{trace}(A_i^T X) \leq b_i \\ & X \text{ positive semidefinite} \end{aligned}$$

Interior point methods not just for continuous optimization

Let G be a graph. A *matching* M is a set of edges without common endvertices.

MAXIMALMATCHING

$$\begin{aligned} \max \quad & \sum_e x_e \\ \text{for all } v : \quad & \sum_{v \sim e} x_e \leq 1 \\ & x \in \{0, 1\}^{E(G)} \end{aligned}$$

LIN-RELAX MAXIMALMATCHING

$$\begin{aligned} \max \quad & \sum_e x_e \\ \text{for all } v : \quad & \sum_{v \sim e} x_e \leq 1 \\ & x \in [0, 1]^{E(G)} \end{aligned}$$